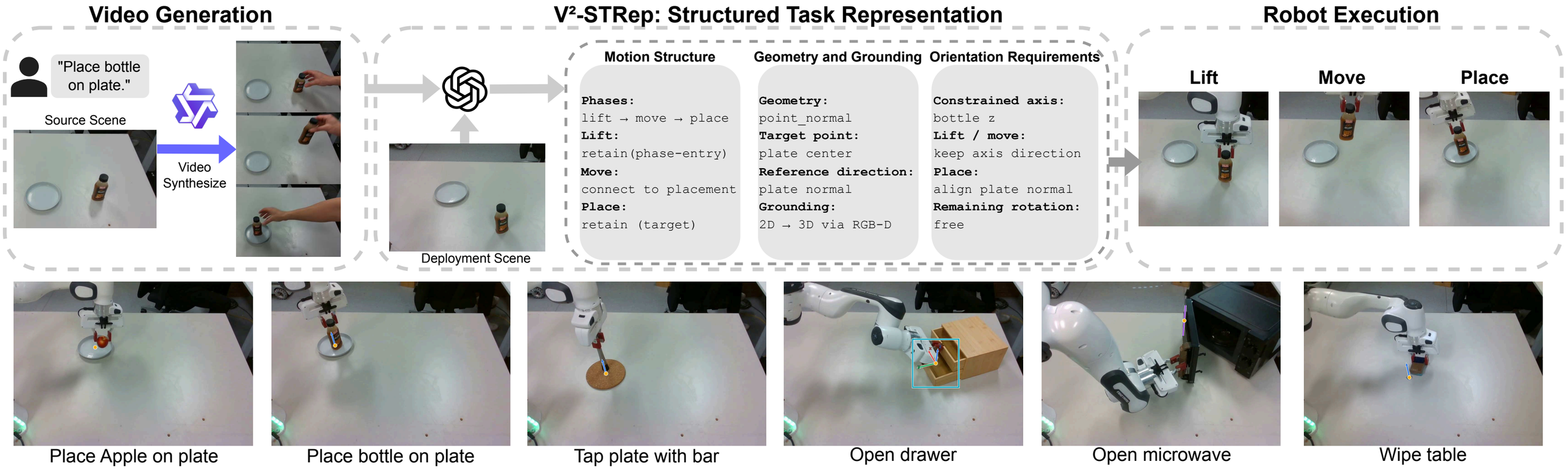

Video Generation
V²-STRep: Structured Task Representation
Robot Execution
"Place bottle on plate."
Source Scene
Video Synthesize
Deployment Scene
Motion Structure
Phases:
lift → move → place
Lift:
retain(phase-entry)
Move:
connect to placement
Place:
retain (target)
Geometry and Grounding
Geometry:
point_normal
Target point:
plate center
Reference direction:
plate normal
Grounding:
2D → 3D via RGB-D
Orientation Requirements
Constrained axis:
bottle z
Lift / move:
keep axis direction
Place:
align plate normal
Remaining rotation:
free
Lift
Move
Place
Place Apple on plate
Place bottle on plate
Tap plate with bar
Open drawer
Open microwave
Wipe table